

# HAMZA SHOUKAT

## Python AI Engineer

+92-341-5535091 / hamzashoukat091@gmail.com / 81/C Agrics Town Phase 1, Raiwind Road, Lahore, Pakistan

---

## PROFILE

Senior AI Python Developer with 4+ years specializing in intelligent automation, SaaS platforms, and enterprise AI solutions. Proven track record of architecting scalable AI systems that serve 10K+ users, reducing operational costs by 40%, and implementing technologies including Claude 4, GPT-5, and Advanced RAG systems. Expert in building end-to-end AI platforms with seamless third-party integrations and autonomous decision-making capabilities.

---

## TECHNICAL SKILLS

- Python, LangChain, Claude 4, GPT-5, Advanced RAG, AI Agents, Agentic Retrieval
  - FastAPI, SQLAlchemy, PostgreSQL, Background Task Processing
  - Web Scraping, SaaS Architecture, Third-party APIs
  - Machine Learning, Data Science, Selenium, Scrapy
  - AWS SageMaker, EC2, Model Deployment, Scalable Architecture
- 

## WORK EXPERIENCE

Senior AI Python Developer

JULY 2024 – PRESENT

*Proximate Solutions, Lahore, Pakistan*

Leading AI development initiatives with focus on intelligent automation and SaaS solutions. Key responsibilities include:

- Architected and developed PerfectCSR, a comprehensive AI-powered customer service platform with advanced bot management, conversation handling, and agentic decision-making capabilities.
- Implemented cutting-edge AI models including Claude 4 and GPT-4.5 with enhanced RAG systems for context-aware responses and intelligent query processing.
- Built robust Agent Controller and Framework systems enabling autonomous decision-making, tool selection, and workflow automation for complex business processes.
- Developed universal web crawling systems and automated data collection pipelines supporting multi-platform integrations including WooCommerce, Shopify, BigCommerce, and messaging platforms.
- Created scalable FastAPI-based architecture with PostgreSQL integration, background task processing, and comprehensive analytics for tracking user engagement and system performance.
- Engineered AI GUI interfaces and automation scripts, delivering seamless user experiences while maintaining enterprise-grade security and reliability standards.

Full Stack Developer | Data Engineer

JULY 2023 – JULY 2024

*Daria Technologies / NIC LUMS, DHA, Lahore, Pakistan*

Working as a Full stack developer. As a part of the Sofizar development team, my key responsibilities are:

- Implemented advanced LLM workflows with chains, agents, and function calling to allow highly sophisticated automation of complex tasks.
- Deployed various models like Llama3-8b, Falcon, Claude 3, and e5-Large on AWS Sagemaker, utilizing data engineering techniques to manage and preprocess input data effectively.
- Created smart LLM knowledge based chatbots and LLM SQL based chatbots with deep knowledge to assist users effectively.
- Prompt Engineering for continuous improvement of systems for better results.
- Spearheaded the creation of a comprehensive logistics platform, seamlessly managing tracking and order booking processes. Ensuring a user-friendly interface for user experiences.
- Developed automated reporting mechanisms for timely and accurate financial analysis.
- Wrote robust APIs to connect front-end and back-end functionalities, ensuring smooth communication and optimal system performance.

Python AI/ML

JUNE 2022 – JULY 2023

*ILSA INTERACTIVE / 455-G3, Johar Town, Lahore, Pakistan*

Worked as a Python and Machine learning Engineer. As a part of the AI FAYVO team, my key responsibilities were:

- Utilized web scraping techniques to gather data from diverse platforms for Machine Learning models.
- Implemented seamless integration of ML models into Django applications.
- Improved adult content detection and categorization by leveraging Random Forest with Google API labels and explicit label extraction.
- Written Python scripts to clean and preprocess image and video datasets.
- Conducted in-depth research to develop and implement novel techniques that boost accuracy.
- Utilized the Matrix Factorization technique to create effective item recommendation systems.
- Prepared and optimized the GPT text-embedding-ada-002 Model for efficient item recommendations.

Python Developer

JUNE 2021 – MAY 2022

*SAW DEVELOPERS / 28K, DHA Phase 1, Lahore, Pakistan*

Worked as a back-end and front-end developer on a multi-vendor e-commerce website. I was part of a small team. My key responsibilities were:

- Effectively organized media and static directories using Django's Model-View-Template architecture.
- Developed and maintained the backend functionalities for admin, seller, and customer portals.
- Designed and built multiple pages using HTML, Bootstrap, JQuery, and Ajax to create a seamless user experience.
- Ensured smooth integration between the frontend and backend components, providing a cohesive and responsive application.
- Managed the codebase using GitHub for version control and deployed the application on an EC2 instance in AWS.

---

## EDUCATION

SUPERIOR UNIVERSITY

JUNE 2022

*BS in Software Engineering*

*Lahore, Pakistan*

---

## CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Generative AI LLMs - NVIDIA-Certified Associate (In Progress)
- Data Cleaning Certification - Kaggle (December 2024)
- Python Programming Certification - Kaggle (April 2024)
- Advanced AI/ML Specialization - Self-directed learning (2023-2024)